

As a Data Analyst on Snowflake's Marketing Analytics team, your objective is to analyze customer relationship management (CRM) data and identify contacts that satisfy two conditions:

1. Contacts who had a marketing touch for three or more consecutive weeks.
2. Contacts who had at least one marketing touch of the type 'trial_request'.

Marketing touches, also known as touch points, represent the interactions or points of contact between a brand and its customers.

Your goal is to generate a list of email addresses for these contacts.

`marketing_touches` Table:

Column Name	Type
event_id	integer
contact_id	integer
event_type	string ('webinar', 'conference_registration', 'trial_request')
event_date	date

`marketing_touches` Example Input:

event_id	contact_id	event_type	event_date
1	1	webinar	4/17/2022
2	1	trial_request	4/23/2022
3	1	whitepaper_download	4/30/2022

4	2	handson_lab	4/19/2022
5	2	trial_request	4/23/2022
6	2	conference_registration	4/24/2022
7	3	whitepaper_download	4/30/2022
8	4	trial_request	4/30/2022
9	4	webinar	5/14/2022

crm_contacts Table:

Column Name	Type
contact_id	integer
email	string

crm_contacts Example Input:

contact_id	email
1	andy.markus@att.net
2	rajan.bhatt@capitalone.com
3	lissa_rogers@jetblue.com

4	kevinliu@square.com
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Example Output:

email
andy.markus@att.net

Solution/Methodology:

1. Approach 1: (Solved using Approach 1)
 - a. YearWeek because Datetrunc does not work in MySQL
 - b. Rank for consecutive weeks
 - c. User and Group for 3 weeks consecutive week identification
2. Approach 2:
 - a. Lead and Lag for find previous and upcoming one
 - b. And then lead = curr -1
 - c. And curr+1 = lag

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with revised_view AS (SELECT
contact_id,
event_type,
yearweek(event_date,3) as yearweek,
rank() over(partition by contact_id order by yearweek(event_date,3)) as rnk
FROM marketing_touches ),
grp_per_view AS (
select
*,
yearweek - rnk as grp
from revised_view )
select
email
from grp_per_view g
join crm_contacts c
on g.contact_id = c.contact_id
WHERE g.contact_id IN(SELECT contact_id //Condition 2
FROM marketing_touches
WHERE event_type = 'trial_request')
group by g.contact_id, grp

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having count(distinct g.yearweek) = 3 //Condition 1
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